

1. Administrative & Study Identification

Full Study Title: A Phase 3 Study to Evaluate the Safety and Efficacy of KarXT + KarX-EC for the Treatment of Agitation Associated With Alzheimer's Disease (ADAGIO-2) **Short Title/ Acronym:** ADAGIO-2 **Protocol Number:** CN012-0024 **NCT Number:** NCT07011745 **Trial Status:** Recruiting **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** KarXT (Xanomeline/Trospium Chloride Capsule) + KarX-EC (Xanomeline Enteric Capsule) **Phase of Development:** Phase 3 **Medical Monitor:** Bristol-Myers Squibb Clinical Operations Lead / Rosalinda Sepulveda, MD, PhD (Medical Director of Clinical Development)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of KarXT + KarX-EC compared to placebo in reducing agitation associated with Alzheimer's Disease. **Secondary Objectives:** To evaluate changes in Clinical Global Impressions-Severity (CGI-S) scores, CMAI-IPA Domains, CMAI Total Score, and to assess the overall safety and tolerability of the combination treatment.

2.2 Background & Rationale To address the significant unmet need for treatments targeting agitation in Alzheimer's Disease (AD), a neuropsychiatric complication affecting approximately half of all AD patients. Currently, treatment options remain extremely limited with only one FDA-approved therapy available. KarXT acts via muscarinic receptor agonism (M1/M4-preferring), offering a novel therapeutic mechanism that influences cholinergic and dopaminergic signaling to safely calm behavioral symptoms, without the typical profile of dopamine receptor blockade.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized **Design:** Parallel Group Study

3.2 Planned Enrollment & Duration Target \$N\$: 352 participants **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 16-Jul-2025 **Estimated Primary Completion:** 03-Nov-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 55 to 90 years. 2. A diagnosis of Alzheimer's disease (AD) in accordance with the 2024 Alzheimer's Association criteria. 3. Confirmation of AD pathology (historical evidence with amyloid PET, A β 42/40 ratio in CSF, pTau181/A β 42 ratio in CSF, or pTau217/A β 42 ratio in plasma using an authorized diagnostic assay; or assessed during eligibility). 4. Mini-Mental State Examination (MMSE) score of 5 to 22, inclusive, at Screening (Visit 1). 5. Have one identified caregiver who has sufficient contact (approximately 10 hours a week or more) and is willing to attend all visits/report on status, oversee compliance, participate in assessments, and provide informed consent. 6. History of agitation that meets the International Psychogeriatric Association (IPA) consensus definition for agitation in cognitive disorders with onset at least two weeks prior to Screening (Visit 1). 7. NPI/NPI-NH Agitation/Aggression score ≥ 4 at Screening (Visit 1) and Baseline (Visit 2). 8. CGI-S ≥ 4 , as related to agitation, at Screening (Visit 1) and Baseline (Visit 2). 9. CMAI-IPA Physical/Verbal Aggression Positivity established at Screening and Baseline (at least 1 of the following 3 criteria): 1 or more aggressive behaviors occurring several times per week; 2 or more aggressive behaviors occurring once or twice per week; or 3 or more aggressive behaviors occurring less than once per week.

4.2 Exclusion Criteria 1. Agitation symptoms that are primarily attributable to a condition other than the AD causing the dementia. 2. History of bipolar disorder, schizophrenia, or schizoaffective disorder. 3. History of (or at high risk for) urinary retention, gastric retention, or narrow-angle glaucoma as evaluated by the Investigator. 4. Risk of suicidal behavior during the study as determined by the Investigator's clinical assessment and/or C-SSR.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: KarXT (Xanomeline/Trospium Chloride [Sanctura]) capsules + KarX-EC (Xanomeline Enteric) capsules vs. Placebo. **Route of Administration:** Oral **Duration of Treatment:** 14 weeks of active treatment.

5.2 Concomitant & Prohibited Therapy Permitted: Selective serotonin reuptake inhibitors (SSRIs) and serotonin norepinephrine reuptake inhibitors (SNRIs) taken at a stable dose for ≥ 8 weeks prior to Screening (Visit 1). Mirtazapine or trazodone used as a hypnotic (if started ≥ 8 weeks prior to Screening). As-needed anxiolytics (e.g., lorazepam). **Prohibited:** Recent history of receiving monoamine oxidase inhibitors, anticonvulsants (e.g., lamotrigine, divalproex), mood stabilizers (e.g., lithium), tricyclic antidepressants (e.g., imipramine,

desipramine), or other psychoactive medications except for as-needed anxiolytics.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | 2 to 4 weeks | Informed Consent, Biomarker Confirmation, MMSE, Medication Washout Review | |
Baseline | Day 0 | Randomization, First Dose, Baseline CMAI-IPA, NPI/NPI-NH, and CGI-S Scoring | | **Interim** | Weeks 1 - 13 | Safety Labs, Efficacy Assessment (Agitation Scales), Caregiver Feedback | |
End of Study | Week 14 | Final Vital Signs, Exit Interview, IP Accountability, Final Efficacy Scoring | | **Follow-Up** | 4 Weeks Post-Tx | Off-treatment safety evaluation and monitoring |

(Note: Timeframes adapted from 14-week treatment duration and screening/follow-up windows)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Mean change from baseline on the Cohen-Mansfield Agitation Inventory-International Psychogeriatric Association (CMAI-IPA) total score at Week 14. **Measurement Method:** Clinician/Investigator assessment via established CMAI-IPA criteria rating scale.

7.2 Secondary & Exploratory Endpoints 1. Mean change from baseline on the Clinical Global Impressions-Severity (CGI-S) at Week 14. 2. Mean change from baseline on the CMAI-IPA Domains at Week 14. 3. Mean change from baseline on the CMAI Total Score at Week 14. 4. Overall safety and tolerability profiling (monitoring for mild-to-moderate side effects such as dry mouth and constipation).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via caregiver interviews and HCP evaluations, with specific attention to anticholinergic/cholinergic side effects (e.g., dry mouth, constipation, urinary/gastric retention). **Serious Adverse Events (SAEs):** Standard protocol 24-hour reporting timeline. **Data Monitoring Committee (DMC):** External independent safety committee utilized to monitor Phase 3 phase trial safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy (CMAI-IPA change); Safety Set for AE/SAE monitoring. **Sample Size**

Justification: Enrolling 352 participants powered appropriately to detect statistically significant reductions in agitation score metrics. **Handling of Missing Data:** Mixed-effect Model Repeated Measure (MMRM) or Multiple Imputation (MI) depending on the variable constraints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring visits by clinical research associates (CRAs) to ensure integrity of caregiver reporting and scale administrations. **Audit Trail:** eCRF locking and data clarification forms via standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: CN012-0024 **Registry Number:** NCT07011745 **Posting ID:** 928706902822789653 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** ADAGIO-2 **Official Regulatory Title:** A Phase 3, Randomized, Double-Blind, Placebo-Controlled, Parallel Group Study to Evaluate the Safety and Efficacy of KarXT + KarX-EC for the Treatment of Agitation Associated With Alzheimer's Disease

12. Clinical Framework (Alzheimer's Disease Specific)

Primary Condition: Agitation Associated With Alzheimer's Disease (Agitation, Alzheimer's Type) **Diagnosis Threshold:** Biomarker-confirmed AD pathology with persistent agitation ≥ 2 weeks, MMSE ≤ 22 , NPI/NPI-NH Agitation ≥ 4 , CGI-S ≥ 4 . **Medical Methodology:** Muscarinic receptor agonism (M1/M4-preferring) via xanomeline-trospium **Optimization Target:** Reduction in behavioral symptoms and CMAI-IPA total score

13. Study Procedures & Methodology

Management Pathway: Structured neuropsychiatric clinical pathway for Alzheimer's dementia management. **Education Protocol (HCP):** Calibration and standardized training for objective administration of the NPI, CGI-S, and CMAI-IPA criteria.

Education Protocol (Patient): Requirement of a dedicated caregiver (available ≥ 10 hours/week) to assist with study compliance and symptom reporting. **Quality Audit Mechanism:** Cross-verification of scale ratings and caregiver perception tracking.

14. Observation & Follow-Up Schedule

Total Duration: Up to 22 weeks (up to 4 weeks screening + 14 weeks treatment + 4 weeks follow-up) **Onsite Visit Cadence:** Baseline, regular intervals during the 14-week treatment phase, and End of Study. **Remote Monitoring Frequency:** Intermittent phone or digital check-ins for safety. **Checklist Review Interval:** Scheduled scale assessments (CMAI-IPA, CGI-S) at designated week markers.

15. Sign-off & Approval

Role	Name	Signature	Date						
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]						
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]						
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]						